Lucky Immanuel Sitanggang

Software Developer

+62 831-8156-8152 | immanuelsgt@gmail.com | linkedin.com/in/luckyimmanuel | github.com/Youngstg
Bukittinggi, West Sumatra | https://luckystg.vercel.app

SUMMARY

Informatics Engineering graduate of Institut Teknologi Sumatera (GPA: 3.77) who cares about building technology people can genuinely use, especially for communities that are often overlooked, such as children with cerebral palsy and the deaf community. Shaped by teaching, lab assistance, and leading student organizations, I work in a collaborative, dependable, and curious way. Seeking a Software Developer role where I can keep learning and create meaningful impact.

EDUCATION

Institut Teknologi Sumatera (ITERA)

Aug 2022 – Jun 2026

Informatics Engineering | GPA: 3.77

WORK EXPERIENCE

Mobile Developer Intern

Jun 2025 – Aug 2025

Karla Bionic

- Developed a cross-platform AAC (Augmentative and Alternative Communication) mobile app for children with Level 3 Cerebral Palsy using React Native (Expo) and Expo Router.
- Designed an inclusive landscape interface with large-scale communication buttons, and a real-time parent monitoring system with push notifications.
- Implemented Firebase Authentication with role-based access (Parent/Child), Firestore data storage, and database security rules; configured EAS builds for app deployment.

Computer Network Lab Assistant

Sep 2025 – Dec 2025

Institut Teknologi Sumatera (ITERA)

- Guided students in configuring VLANs, DHCP, and Wireless Access Points using Cisco technologies.
- Assisted students in implementing and configuring mail servers using SMTP and Postfix in a networking laboratory environment.

Teaching Assistant | Formal Languages and Automata Theory

Sep 2024 – Dec 2024

Institut Teknologi Sumatera (ITERA)

- Acted as liaison between students and lecturers, managing substitute class schedules and assessments.
- Conducted targeted tutoring sessions to help students master complex theoretical topics.

Laboratory Assistant PKS 1 & 2

Aug 2023 – Dec 2024

Institut Teknologi Sumatera (ITERA)

- Prepared instructional materials and managed full responsibility for task grading and quizzes for multiple semesters.
- Designed evaluative assessments in collaboration with faculty to ensure academic standards were met.

ORGANIZATIONAL EXPERIENCE

Specialist Staff | Web & IT Development

Feb 2025 – Dec 2025

Keluarga Mahasiswa ITERA (KM ITERA), student union

- Optimized university and KM ITERA websites to improve performance, user experience, and digital accessibility.
- Led digitalization of inter-organization activities by developing IT-based products with Laravel, MySQL, and REST APIs, using Git for version control, to drive campus innovation.

Staff PPK

Aug 2023 – Dec 2024

Himpunan Mahasiswa Informatika (HMIF), Informatics student association

- Led the Thesis Expo as Head of Event, managing all operations from planning to execution.
- Coordinated the FTI Expo and arranged seminars featuring expert speakers for Informatics Engineering.

PROJECTS

CareCircle | Family Care Coordination Web App
2026

- Built a family care coordination web app with Next.js, TypeScript, and Better Auth, featuring role-based access (Owner, Coordinator, Member), task assignment, and activity history.
- Designed a relational schema with Prisma ORM and MySQL, including migrations, unique memberships, and indexed queries for tasks, documents, and activity.
- Implemented private document storage on MinIO with server-side access checks and upload validation (size, MIME type, file signature); containerized the stack with Docker Compose.
- Added automated tests with Vitest and enforced code quality with ESLint.

Thesis | SIBI Sign Language Recognition
May 2026

- Built a deep learning system to translate SIBI (Indonesian Sign Language) gestures from video, supporting communication accessibility for the deaf community.
- Benchmarked VideoMAE and ViViT transformers via 5-fold cross-validation, best model (fine-tuned ViT-B/16) reached 97.18% accuracy, outperforming ViViT and I3D baselines.

Digo | AI Sales Assistant Chatbot for Digital Agency
digo.us.ci

- Built Digo, an AI sales assistant chatbot for digital agencies, using NestJS for the backend and React 19 for the web chat widget.
- Implemented Google Gemini with retrieval-augmented generation (RAG) over services, pricing, and FAQ data, and integrated Telegram and WhatsApp channels.

KosMatch
kosmatch.us.ci

- Built KosMatch, a Next.js decision support system (DSS) that helps students and workers compare boarding houses objectively, using PostGIS geospatial data and an interactive campus map.
- Calculated true monthly cost by combining rent, utilities (electricity, water, internet, laundry), and estimated daily commuting cost, with a scenario simulator for usage and tariff changes.
- Separated mandatory requirements from flexible preferences and enabled side-by-side comparison of up to 3 options in a single table.

COMPETITIONS

GERAKIN
Jun 2024

- Achieved 3rd place in an internal campus prototype competition, building a computer vision-based game with OpenCV.
- Engineered interactive gameplay to encourage physical exercise in children through real-time motion detection.

CERTIFICATES

AWS Certified Cloud Practitioner
Issued Jul 2026 · Expires Jul 2029

Amazon Web Services (AWS)

TOEFL | ITERA English Proficiency Test (InciTe)
Apr 2026

Institut Teknologi Sumatera Language Center | Total Score: 513

SKILLS

Programming Languages: JavaScript, TypeScript, Python, PHP, C++, SQL

Web Development: React, Next.js, Node.js, NestJS, Laravel, HTML5, CSS3, Tailwind CSS, REST API

Mobile Development: React Native, Firebase

Databases: MySQL, PostgreSQL, PostGIS, Prisma ORM, Firebase (NoSQL)

Computer Vision & AI: OpenCV, Machine Learning, Deep Learning, Image Processing, Google Gemini API, RAG

Tools & Platforms: Git, GitHub, Docker, Vitest, ESLint, AWS (Cloud Practitioner), Cisco